

Classification & Risk Scoring Pipeline

STEP 1: Preprocessing

Raw Sequence
FASTA / plain text

clean_sequence()
Regex filter, min length ≥ 10

Cleaned Sequence
20 canonical amino acids

STEP 2: Features

Feature Extraction
21-D vector per sequence

seq_length
(1)

aa_freq_A...Y
(20)

Feature Vector
[seq_length, aa_freq_*]

STEP 3: Training

80/20 Split
random_state = 42

Logistic
Regression

Random
Forest

F1 Selection
→ best_model.joblib

Calibration

Per-class centroids
 $\mu_c = \text{mean}$, $\sigma_c = \text{std}$
($\sigma_c \geq 1e-8$)

Inference

StandardScaler
predict_proba[:, 1]
threshold ≥ 0.5

Risk Scoring

$\|x - \text{centroid}_c\|_2$
 $z = (\text{dist} - \mu_c) / \sigma_c$
risk = 50 + 15 × z

OUTPUT

**Predicted
Virus**

**Confidence
(probability)**

**Risk
Score**

**Risk
Category**

**Z-score
(atypicality)**

Category Thresholds

Harmless
<20

Neutral
20-39

Moderate
40-59

Dangerous
60-79

Critical
≥80

Deployment: Streamlit app

<https://mutation-analysis.streamlit.app> | streamlit run app.py